Sample Public School

Unit Test: Electricity

Subject: Physics Class: Class 10

Time allowed: 45 minutes **Maximum marks: 20**

TEACHER'S COPY · ANSWER KEY AT THE END

General instructions

1. All questions are compulsory.
2. Section A has 2 multiple choice questions of 1 mark each.
3. Show the formula and working for every numerical question.

SECTION A

Multiple Choice Questions

Choose the correct option. (2 × 1 = 2 marks)

1. The SI unit of electric resistance is: [1]

A) volt B) ampere

C) ohm D) watt

2. Which instrument is used to measure electric current in a circuit? [1]

A) voltmeter B) ammeter

C) galvanometer D) potentiometer

SECTION B

Short Answer Questions

Answer in two or three sentences. (2 × 2 = 4 marks)

3. State Ohm's law. [2]

4. Define electric power and give its SI unit. [2]

SECTION C

Numerical Questions

Show the formula, working and units. (2 × 3 = 6 marks)

5. A 12 V battery is connected across a 4 Ω resistor. Find the current in the circuit and the power used by the resistor. [3]

6. Three resistors of 2 Ω, 3 Ω and 6 Ω are connected (a) in series and (b) in parallel. Find the equivalent resistance in each case. [3]

SECTION D

Long Answer Questions

Answer in detail. (2 × 4 = 8 marks)

7. Why are household appliances connected in parallel and not in series? Give three reasons. [4]

8. Explain the heating effect of electric current. State Joule's law of heating and give two applications. [4]

— End of question paper —

Answer key and marking scheme

Unit Test: Electricity · Physics · Class 10. Use this as the marking scheme; it is also the model answer used for AI grading.

Section A · Multiple choice

1. 1 mark · Moderate

(C) ohm

2. 1 mark · Moderate

(B) ammeter

Section B · Short answer

3. 2 marks · Moderate

- At constant temperature, the current through a conductor is directly proportional to the potential difference across its ends (1 mark).
- Written as V = IR, where R is the resistance (1 mark).

4. 2 marks · Moderate

- Electric power is the rate at which electrical energy is consumed or work is done, P = VI (1 mark).
- Its SI unit is the watt, W (1 mark).

Section C · Numerical problems

5. 3 marks · Moderate

I = V/R = 12/4 = 3 A (1.5 marks).
P = VI = 12 × 3 = 36 W (1.5 marks).

6. 3 marks · Moderate

(a) Series: R = 2 + 3 + 6 = 11 Ω (1.5 marks).
(b) Parallel: 1/R = 1/2 + 1/3 + 1/6 = 1, so R = 1 Ω (1.5 marks).

Section D · Long answer

7. 4 marks · Moderate

- Each appliance gets the full mains voltage of 220 V (about 1.5 marks).
- If one appliance is switched off or fails, the others keep working (about 1.5 marks).
- Each appliance draws current according to its own resistance and can have its own switch and fuse (1 mark).

8. 4 marks · Moderate

- Current through a resistor converts electrical energy into heat (1 mark).
- Joule's law: $H = I^2Rt$, heat is proportional to the square of the current, the resistance and the time (2 marks).
- Two applications, e.g. electric iron, heater or geyser, filament bulb, fuse (1 mark).